

LOGIC No: SEQ-5500	DESCRIPTION: DRYER SYSTEM (YD-2301) SHUTDOWN LOGIC	SIL: SIL 2	TAG: YD-2301 FLOC: TJC-LLD-2300-01
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CAUSE & EFFECT MATRIX

ID	INITIATOR / CAUSE (INPUT)	TAG	SET POINT	VOTE	EFF-1 TRIP DRUM DRIVE MOTOR	EFF-2 CLOSE STEAM INLET TV-2301	EFF-3 TRIP CENTRIFUGE FEED GF-2210	EFF-4 OPEN N2 PURGE FV-2302 (max inert)	EFF-5 ANNUNCIATE ALARM ON DCS
T1	Outlet temperature HIGH-HIGH	TSHH-2301	> 125 degC	1oo1	X	X			X
T2	Nitrogen purge flow LOW-LOW	FSLL-2302	< 200 kg/h	2oo3	X	X		X	X
T3	Inlet chute level HIGH-HIGH	LSHH-2303	> 90 %	1oo1			X		X
T4	Drum speed LOW-LOW (chain slip)	SSLL-2305	< 1 rpm	1oo1		X			X
T5	Vent O2 HIGH-HIGH	ASHH-2307	> 8 % O2	1oo2	X	X		X	X
T6	Drive motor overload	MPR-2301	MPR pickup	1oo1	X	X			X

Legend: X = initiator (row) actuates that effect (column). Voting: 1oo1/1oo2/2oo3 = number of transmitters that must agree to trip.

EFFECT ID	FINAL ELEMENT / ACTION
EFF-1	TRIP DRUM DRIVE MOTOR
EFF-2	CLOSE STEAM INLET TV-2301
EFF-3	TRIP CENTRIFUGE FEED GF-2210
EFF-4	OPEN N2 PURGE FV-2302 (max inert)
EFF-5	ANNUNCIATE ALARM ON DCS

#	START PERMISSIVE (AND-gate)	SIGNAL
1	Nitrogen inerting established (O2 < 5%)	AT-2307
2	Steam condensate drain open	ZSO-2304
3	Lube oil to gearbox OK	PSL-2306
4	No active trip / reset done	DCS reset
=>	ALL permissives TRUE (AND) => START YD-2301 ENABLED	RUN

Notes: 1. Trip set points are DUMMY training values. 2. On any trip the effects marked X are actuated by the Safety PLC and annunciated on the DCS. 3. A trip is latched and requires a manual reset once the cause has cleared and permissives are healthy. 4. Refer to P&ID; TJC-LLD-PID-2301 for instrument loops and the Datasheet for design limits of YD-2301.